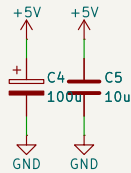
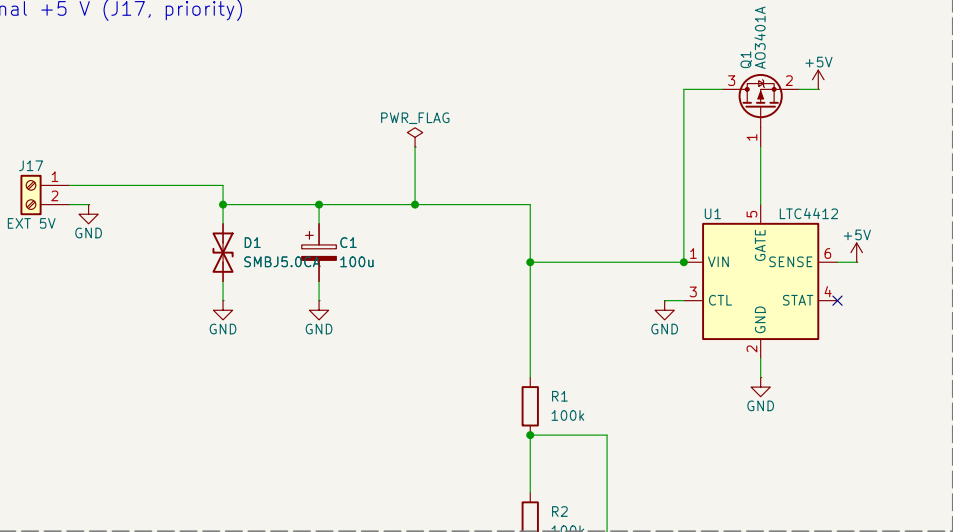


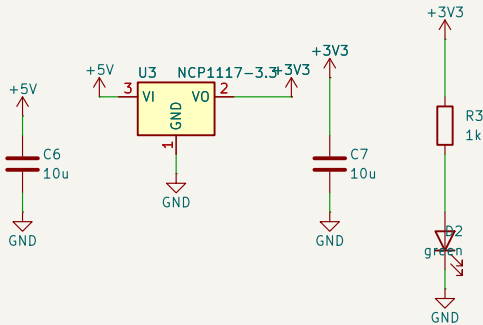
1	2	3	4	5	6	7	8
SAM CPU replacement board, draft 0.2, drawn from docs/ARCHITECTURE.md. Licence: CERN–OHL–S v2. A Raspberry Pi 4 on the 40–pin header J21 runs PinMAME + ppuc (HDMI stays on the Pi). The RP2354B drives the original IO power driver board on J9 and scans the switches. Connector names and pinouts follow the original CPU/Sound board 520–5246–00. Each connector sits on the same page as the circuit that drives or reads it. Signals between pages use global labels; the numbers in brackets next to a label list the other pages it goes to.							
A							
B	Power Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors File: power.kicad_sch		MCU and Pi RP2354B SAM IO controller and Raspberry Pi 4 header File: mcu.kicad_sch		IO bus J9: bus to the IO power driver board (IO board J1) File: io_bus.kicad_sch		
C	Switch columns Switch matrix columns: 74HC595, strobe drivers, J1 File: sw_columns.kicad_sch		Switch rows 1–8 Switch matrix returns 1–8: J6, LM339 front end, 74HC165 File: sw_rows_1.kicad_sch		Switch rows 9–16 Switch matrix returns 9–16: J12, LM339 front end, 74HC165 File: sw_rows_9.kicad_sch		
D	Dedicated switches 1–16 Dedicated switches D1–D16 (J2, J3) and coin door memory protect File: sw_dedicated_1.kicad_sch		Dedicated switches 17–24 Dedicated switches D17–D24 (J13) and DIP switches File: sw_dedicated_2.kicad_sch		GI dimmer J18 GI dimmer header File: display_gi.kicad_sch		
E	Audio Audio: PCM5102A, TDA7297 stereo + subwoofer (+12 V) File: audio.kicad_sch						
F	RS485 and I2S RS485 port (RJ45) and I2S input header File: rs485_i2s.kicad_sch						
			Stern SAM CPU PPUC (CERN–OHL–S v2) Sheet: / File: sam_cpu.kicad_sch Title: SAM CPU replacement board Size: A3Date: 2026–10–06Rev: 0.2 KiCad E.D.A. 10.0.6Id: 1/13				
1	2	3	4	5	6	7	8

J11 keeps the original CPU board connector (IO board J16 harness) for +5 V and ground only: its +−12 V pins are not used.
The audio amplifiers run from an external +12 V supply on J23 (about 3 A at full volume). F1 + D5 blow the fuse on reverse polarity.
U3 NCP1117 (as in the RP2350 hardware guide) is stable with ceramic capacitors. All regulator capacitors are 0805 or larger for low DC−bias derating.
External +5 V on J17 has priority: U2 turns the J11 path off whenever +5V_EXT is present (EXT_PRESENT high).
LTC4412 + AO3401A ideal diodes stand in for the TPS2121 of ARCHITECTURE.md 9.3 (library part, about 4 A per path).
JP1 (bridged by default) disconnects the J11 +5 V completely; leaving J17 unplugged forces J11.

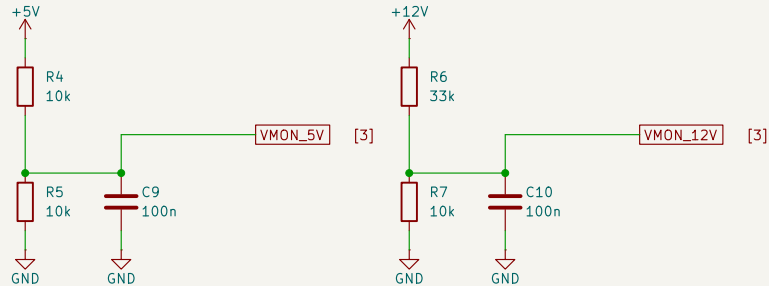
External +5 V (J17, priority)



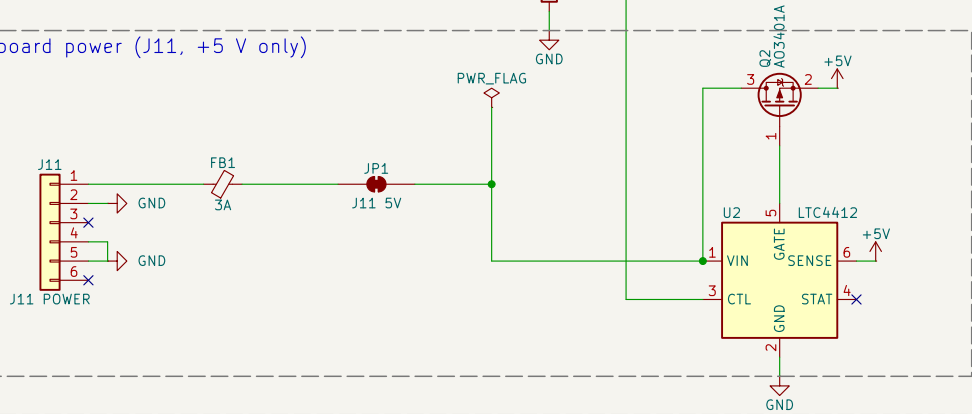
3.3 V



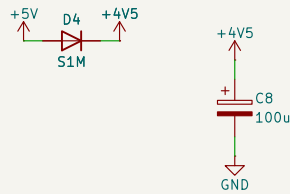
Supply monitors (RP2354B ADC)



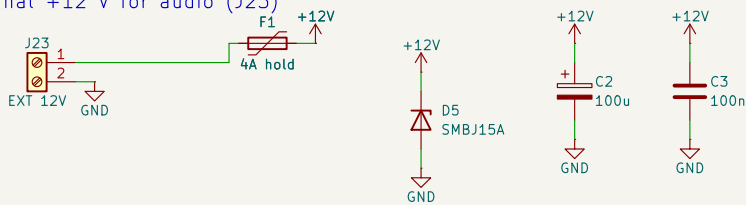
IO board power (J11, +5 V only)



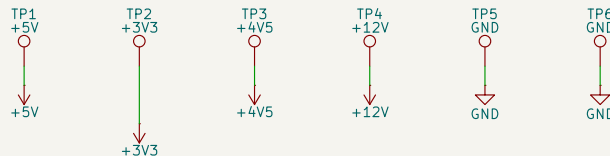
4.5 V for the switch matrix



External +12 V for audio (J23)



Test points



Stern SAM CPU PPUC (CERN−OHL−S v2)

Sheet: /Power/
File: power.kicad_sch

Title: Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors

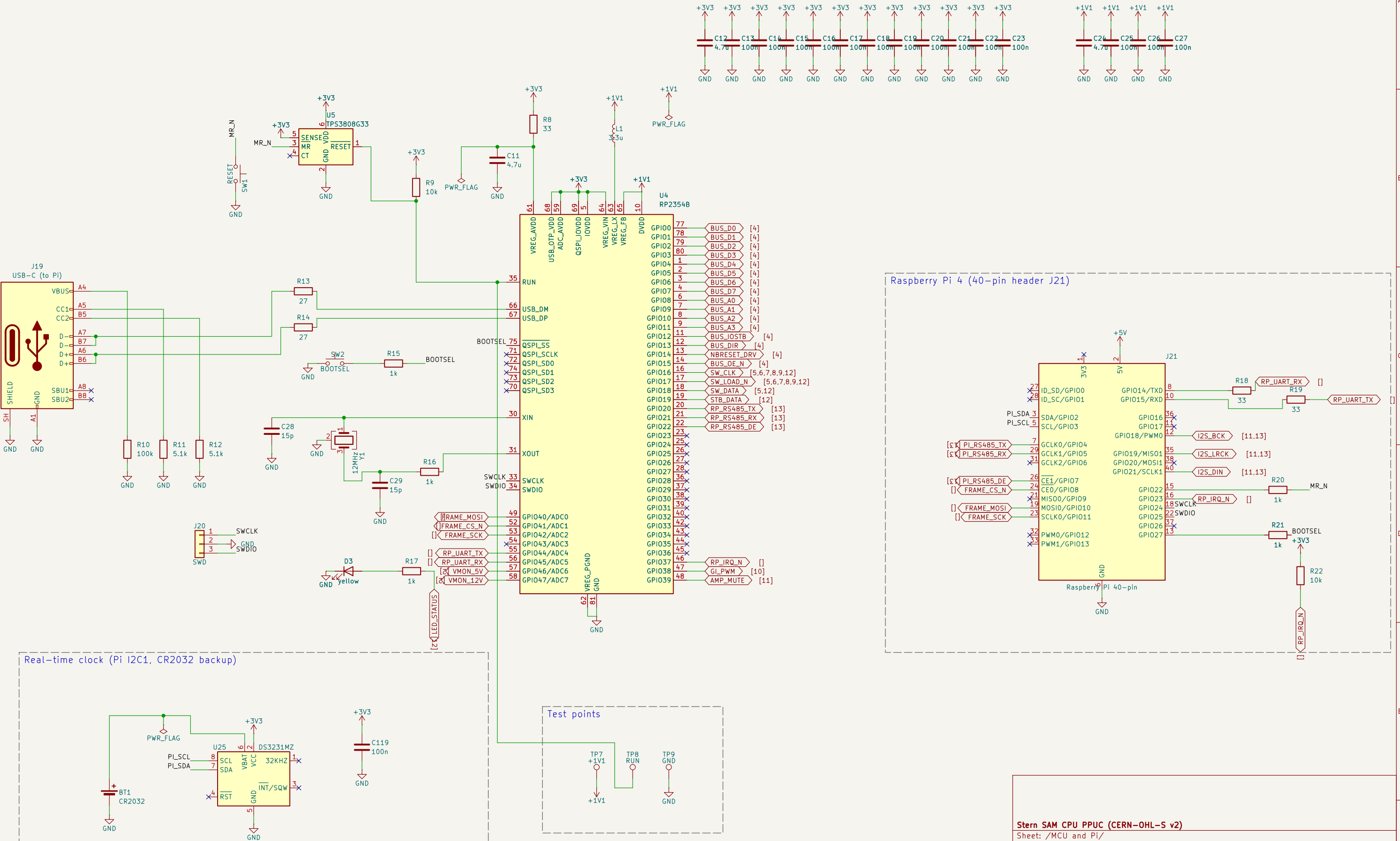
Size: A3 Date: 2026−10−06

KiCad E.D.A. 10.0.6

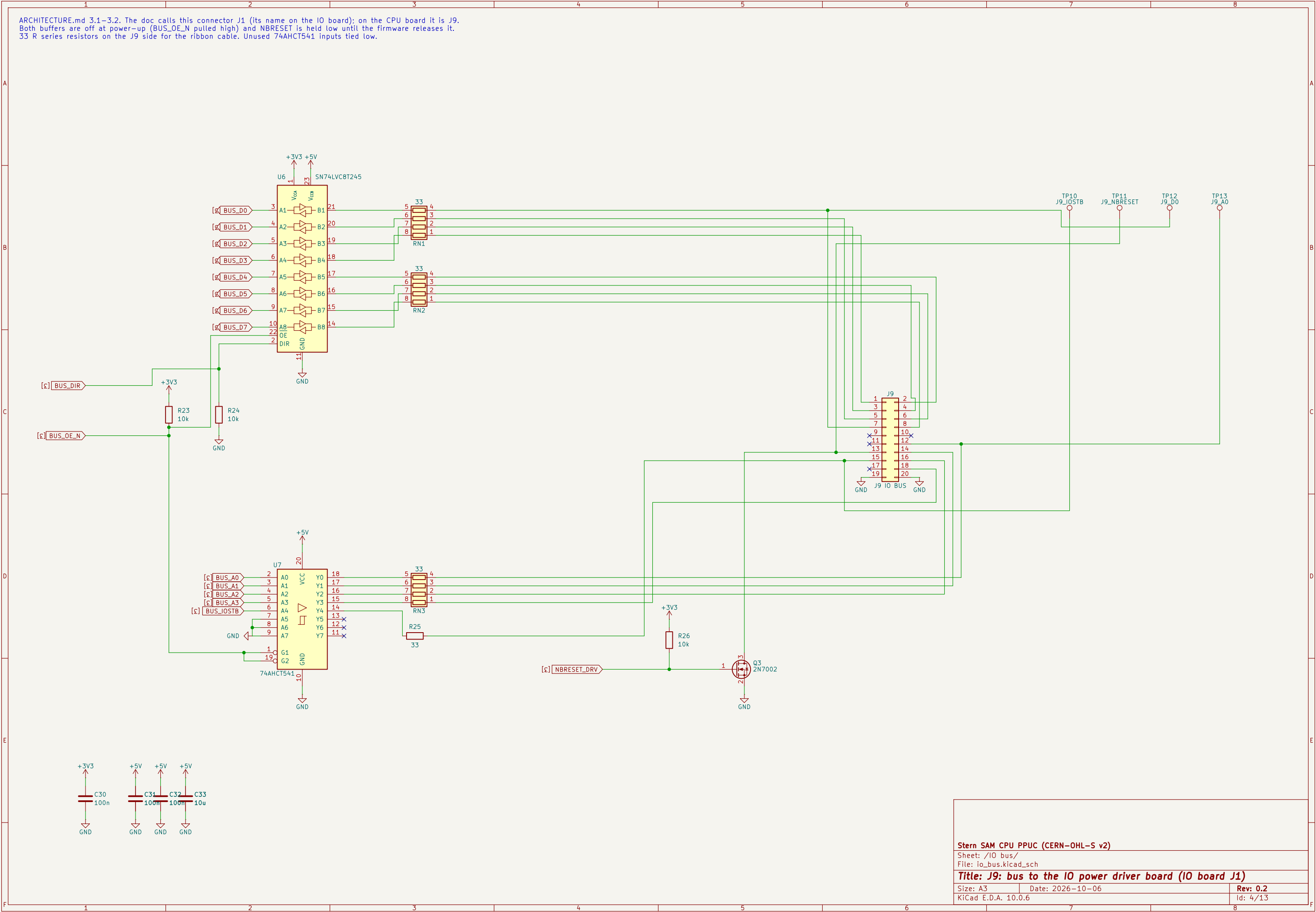
Rev: 0.2

Id: 2/13

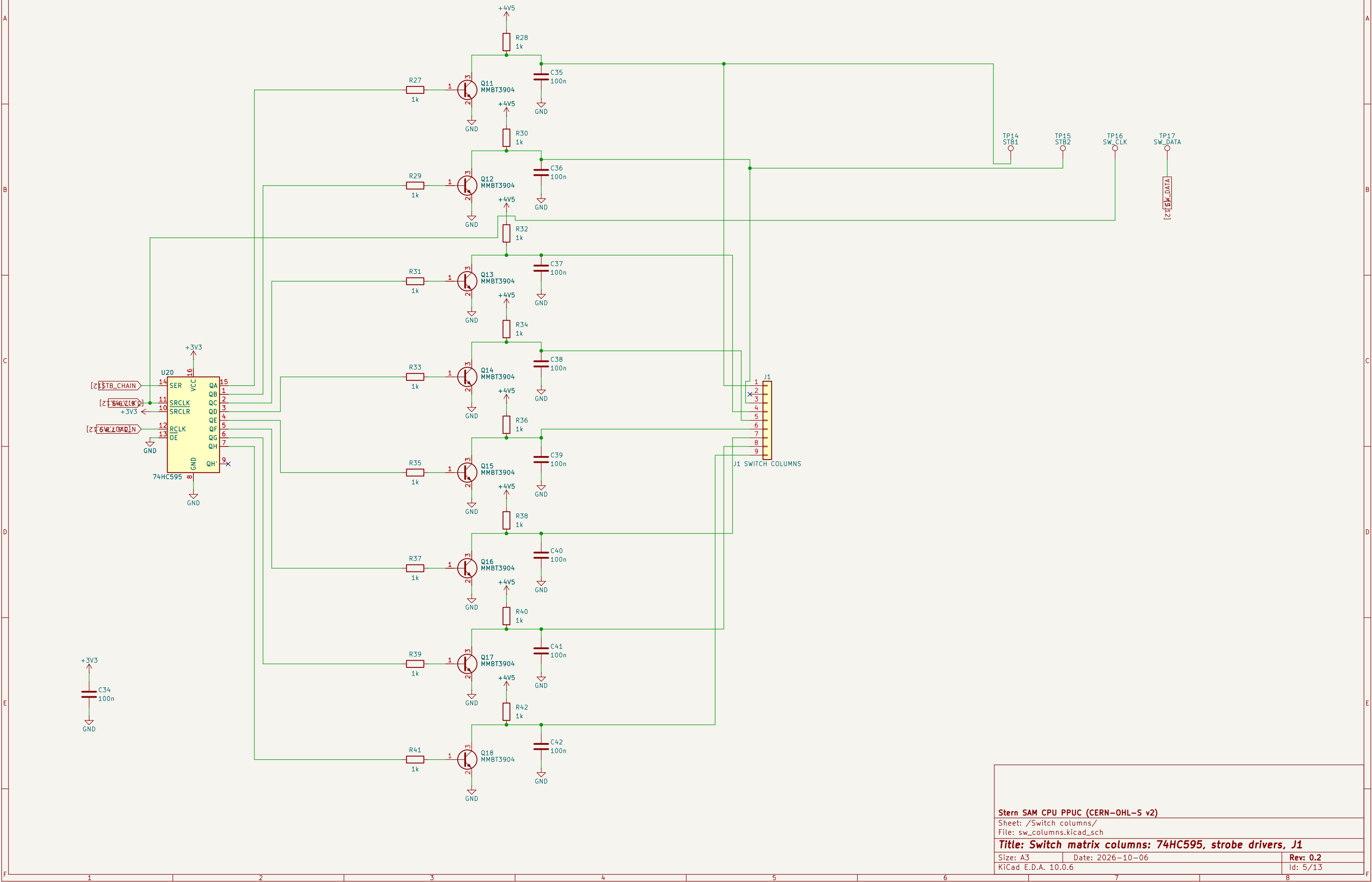
GPIO map: ARCHITECTURE.md 9.2. PIO0 bus on GPIO0–15, PIO1 switch chain on GPIO16–20, GPIO20–22: RS485 port (UART1 TX / RX, driver enable, page 13); GPIO23–36 unassigned. Pi UART3 (GPIO4 / 5, RTS on GPIO7) reaches the same port. Frames from the Pi on SPI1 (GPIO40–42), UART0 on GPIO44–45, ADC monitors on GPIO46–47, GPIO43 spare. Coin door, status LED and GLSPARE moved onto the switch chain. Core supply from the internal switching regulator: VREG_LX → L1 → DVDD (1.1 V); VREG_AVDD through 33 R / 4.7 uF. Check against the RP2350 hardware guide. Raspberry Pi 4 on the 40-pin header J21 (HDMI stays on the Pi). The Pi is powered from the board +5 V; never plug the Pi USB–C supply at the same time. U25 DS3231MZ real-time clock on the Pi I2C1 bus (Linux: dtoverlay=i2c-rtc.ds3231), backed by a CR2032 in BT1.



ARCHITECTURE.md 3.1–3.2. The doc calls this connector J1 (its name on the IO board); on the CPU board it is J9. Both buffers are off at power-up (BUS_OE_N pulled high) and NBRESET is held low until the firmware releases it. 33 R series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.



U20 latch shares SW_LOAD_N with the 74HC165 load (one pulse: 165 loads while low, 595 latches on the rising edge), U20 now follows U31 (DMD panels sheet), which carries the status LED and GI_SPARE.
One shift clock for both registers: PIO1 shifts the next strobe pattern into the 595 while the 74HC165 chain shifts in (ARCHITECTURE.md 5.3–5.4).
Strobes: 595 output high -> MMBT3904 on -> strobe pulled low (active), with the original 1 k pull-up to +4.5 V and 100 nF.
J1 keeps the original pinout (key on pin 2). Strobes 5–8 are driven even where a game does not use them (Q15).



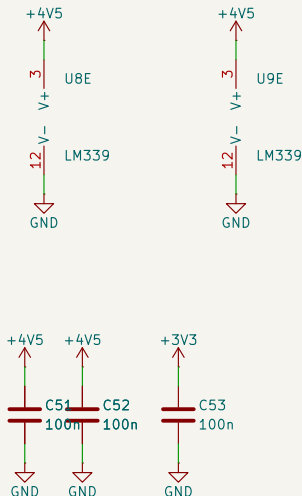
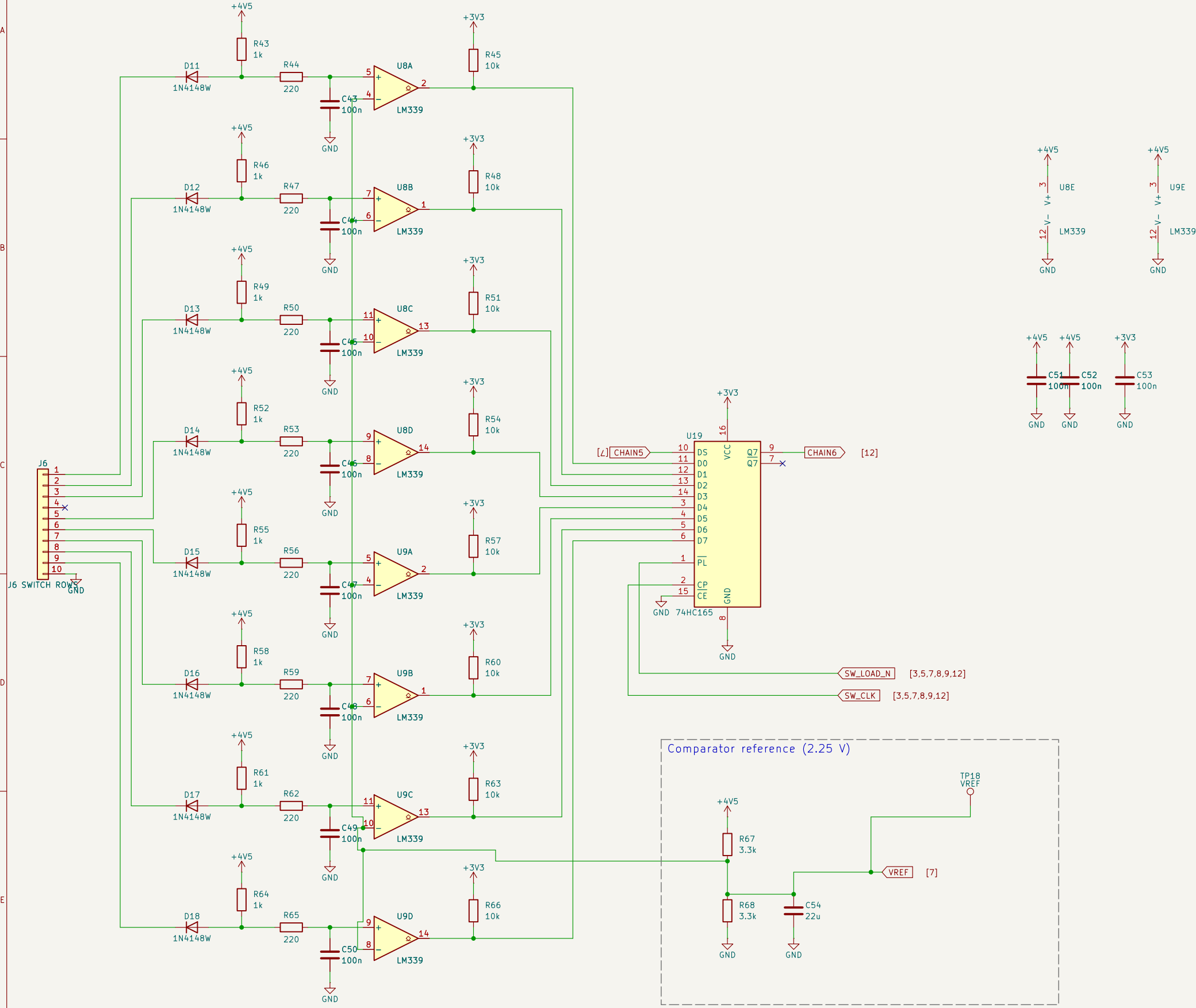
Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Switch columns/
File: sw_columns.kicad_sch

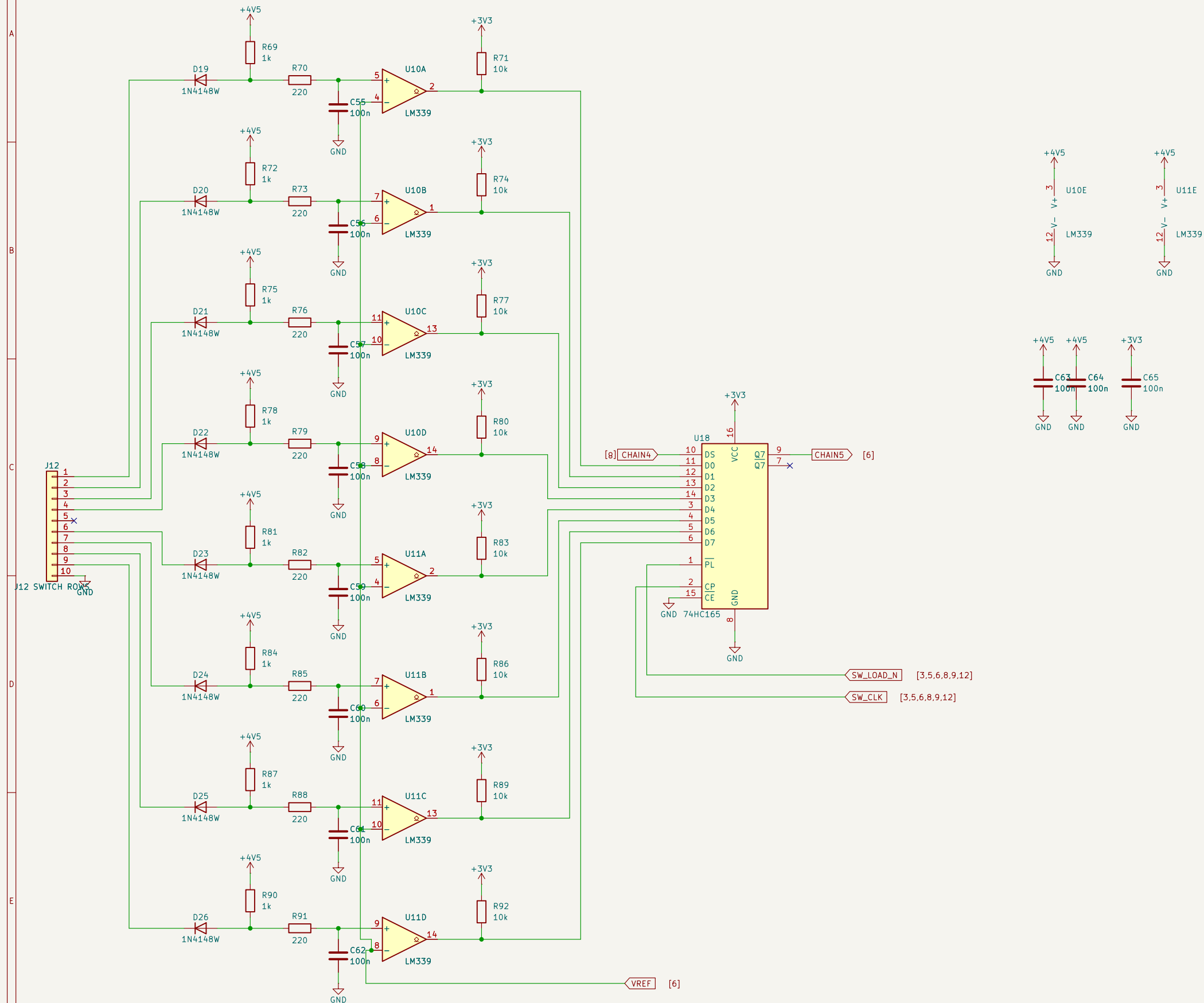
Title: Switch matrix columns: 74HC595, strobe drivers, J1

Size: A3	Date: 2026-10-06	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 5/13

Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.

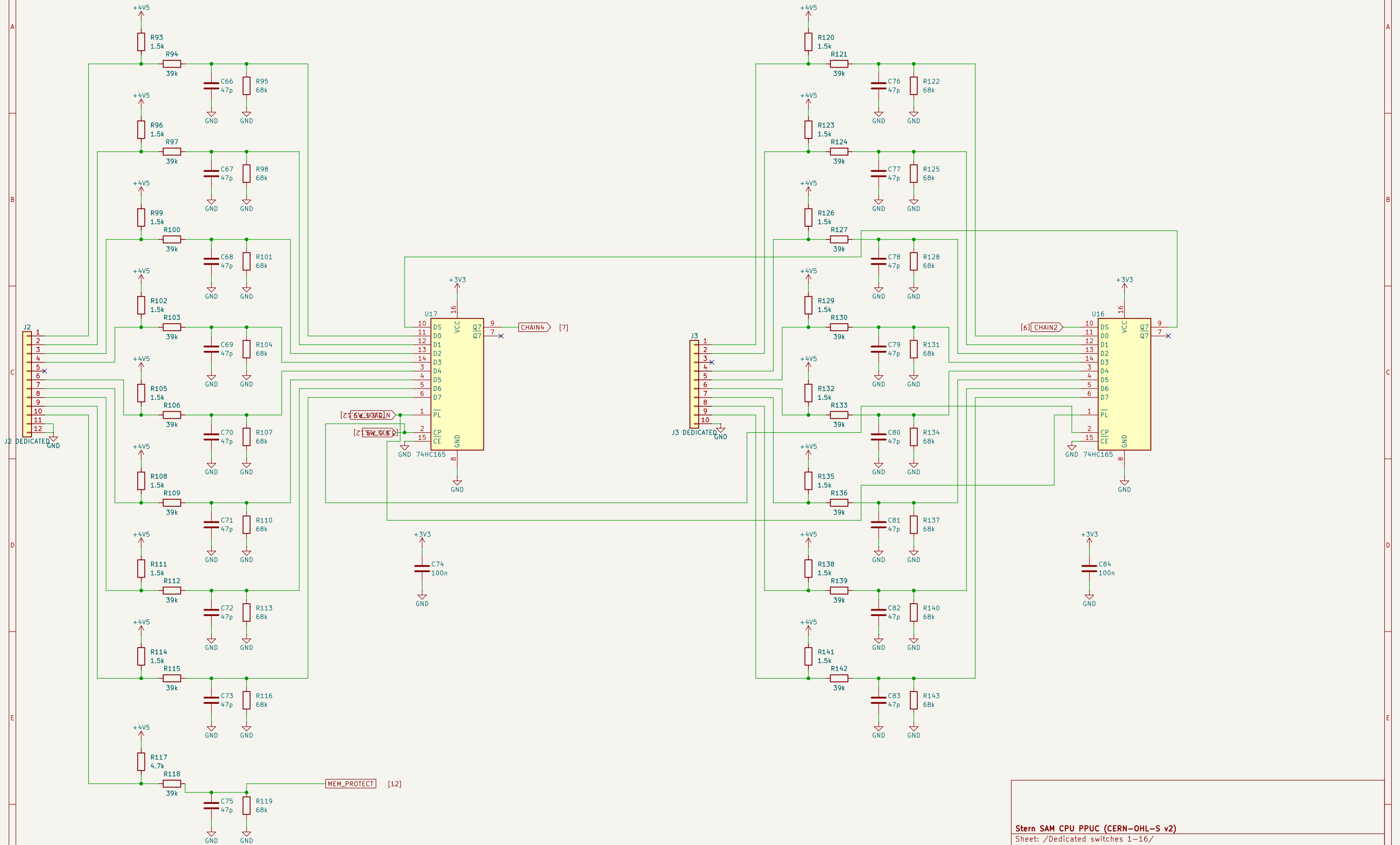


Per return (ARCHITECTURE.md 5.2-5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing; check it with a meter.

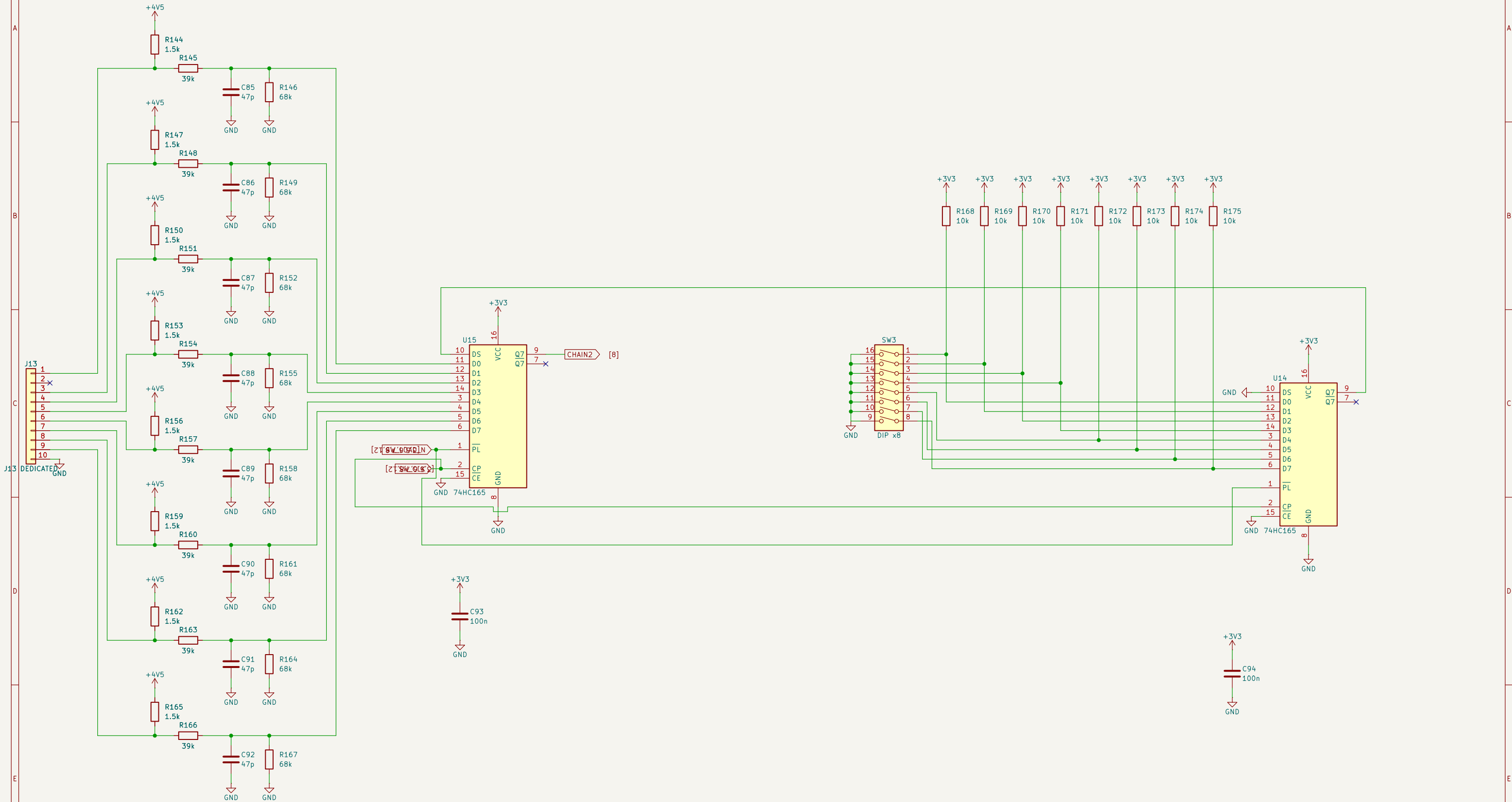


Stern SAM CPU PPUC (CERN-OHL-S v2)
 Sheet: /Switch rows 9–16/
 File: sw_rows_9_kicad_sch
Title: Switch matrix returns 9–16: J12, LM339 front end, 74HC165
 Size: A3 Date: 2026–10–06 Rev: 0.2
 KiCad E.D.A. 10.0.6 Id: 7/13

Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. Memory protect (J2 pin 10) uses the same filter with the original 4.7 k pull-up and goes to input D0 of the 7th 74HC165 (U30, DMD panels sheet).

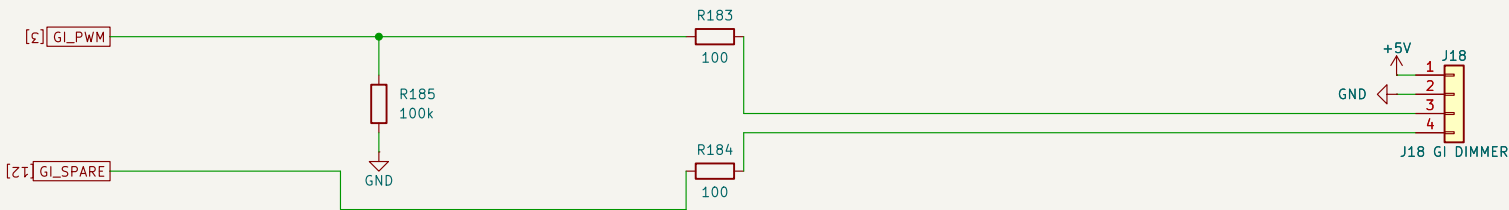


Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. The 8 DIP switches are read through the last 74HC165 of the chain (U14), whose serial input is tied low.

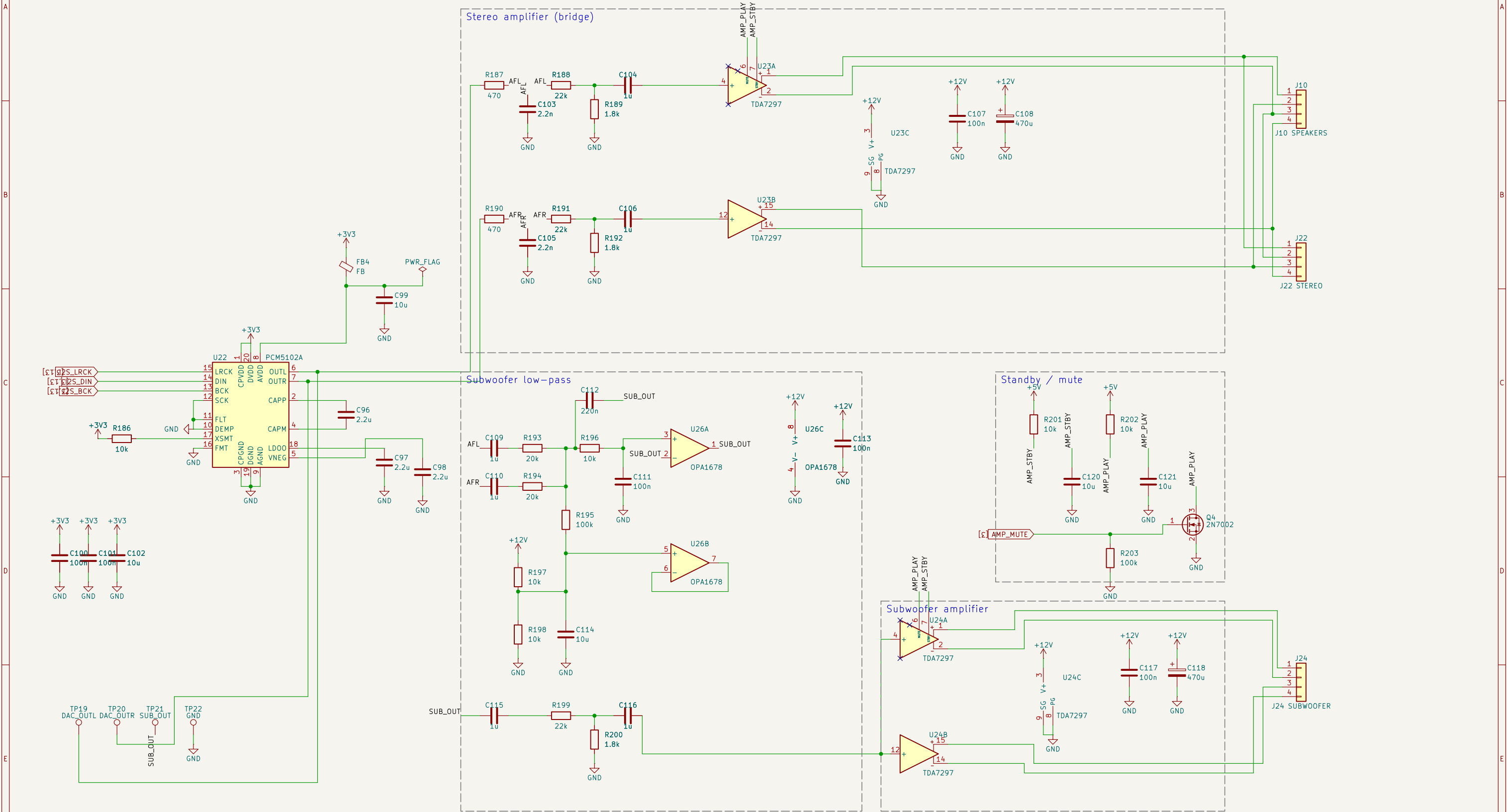


J18 is the default GI dimmer link of ARCHITECTURE.md 10.2: +5 V, GND, GI_PWM, GI_SPARE (GI_SPARE is an on/off output of U31).
The DMD interfaces were removed (2026-10-07): DMD signals do not survive the long cabinet wires.

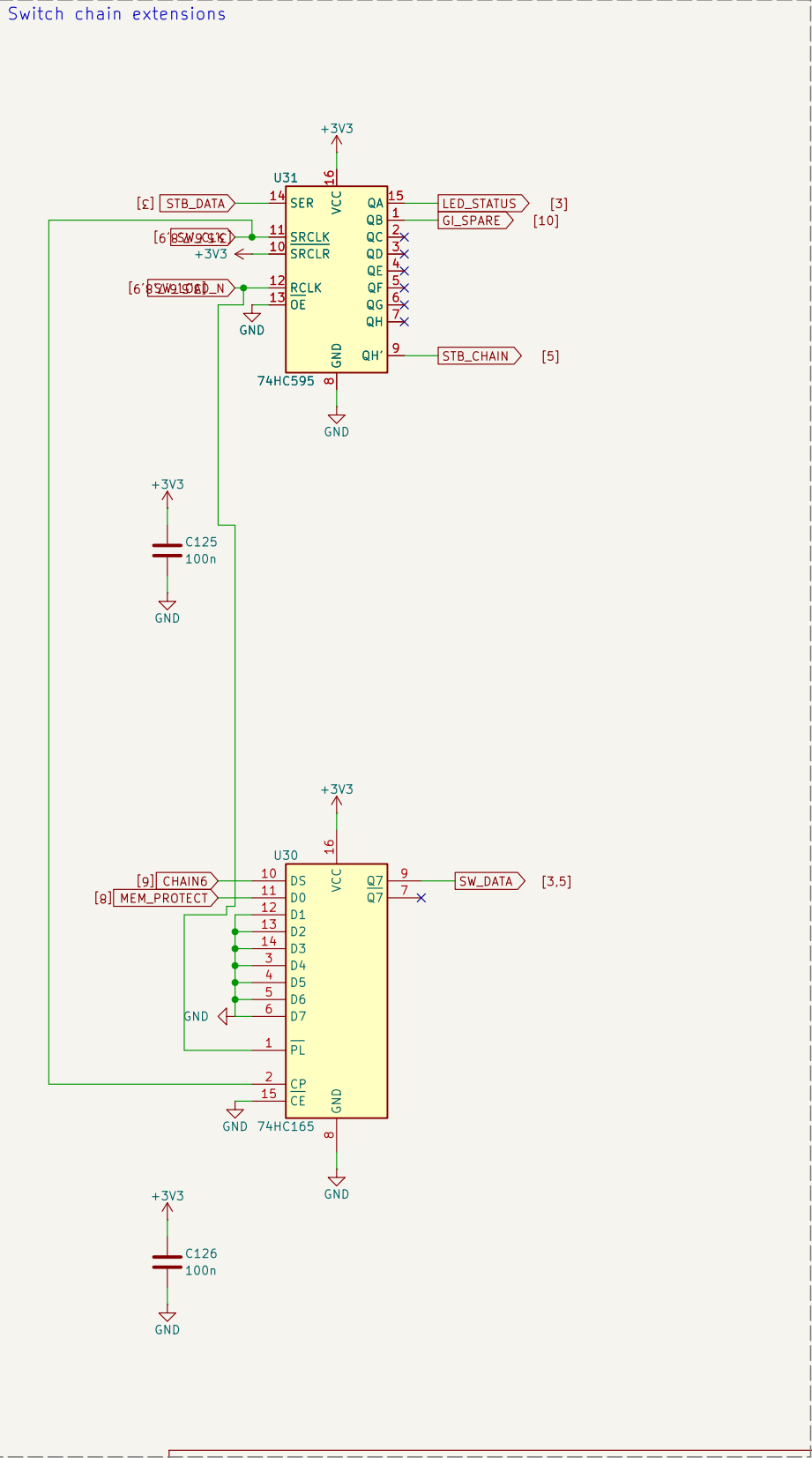
GI dimmer board link (J18)



Single +12 V supply (J23). U23 drives left / right in bridge (BTL): no output capacitor, but no speaker wire may go to ground.
J10 keeps the original connector: pins 1 / 2 = left + / right +, pins 3 / 4 (ground on the original) = left - / right -. Check the harness keeps the two speaker returns separate.
J22 = the same two channels as pairs (L+ L- R+ R-). Subwoofer: (L + R) / 2 through a 110 Hz 2nd-order low-pass (U26). U24 drives two bridged outputs on J24 (dual voice coil or two subwoofers).
TDA7297 gain is fixed at 32 dB: 22k / 1.8k input dividers give full output at DAC full scale. AMP_MUTE (RP2354B GPIO37) high mutes all outputs. U23 / U24 need a heatsink.



U31: first 74HC595 of the strobe chain (status LED, GI_SPARE, 6 spare outputs). U30: last 74HC165 of the switch chain (coin door memory protect + 7 spare inputs). Both share SW_CLK and SW_LOAD_N (165 load / 595 latch on one pulse). The HD DMD panel outputs that shared this page were removed (2026-10-07).



One half-duplex RS485 port on J27 (RJ45: pin 1 A, pin 2 B, pins 7 / 8 GND, DMX-over-RJ45 order), THVD1450: 3.3 V, bus fail-safe, +18 V ESD; SM712 TVS for the cable; 120 Ω termination fitted with the JP2 jumper (end of line only). Pi on J21: Pi UART3 (GPIO4 TX, GPIO5 RX, GPIO7 RTS = driver enable, Linux RS485 mode). Pi remote: RP2354B UART1 (GPIO20 TX, GPIO21 RX, GPIO22 driver enable). Both reach the transceiver through 1 k; only the configured side drives. J28: I2S from a remote Pi to the PCM5102A (in parallel with J21 pins 12 / 35 / 40). Keep that cable short: I2S is clocked logic and has the same long-wire limits as the DMD links that were removed.

